Antiplatelet, anticoagulant, and fibrinolytic activity in vitro of extracts from selected fruits and vegetables.

A diet rich in fruits and vegetables is known to decrease the risk of cardiovascular disease. However, the information regarding the antithrombotic activity (antiplatelet, anticoagulant, and fibrinolytic) of fruits and vegetables is scarce. The aim of this study was to assess the antithrombotic activity of extracts from fruits and vegetables widely consumed in central Chile. The study included samples of 19 fruits and 26 vegetables, representative of the local diet. The extracts prepared from each sample included an aqueous (juice or pressed solubles) and/or methanol-soluble fraction. The extracts were evaluated for antiplatelet, anticoagulant, and fibrinolytic activity in vitro at a final concentration of 1 mg/ml. The antiplatelet activity was assessed by platelet aggregation inhibition; anticoagulant activity was measured by the prothrombin time (PT), diluted prothrombin time (dPT), activated partial thromboplastin time (APTT), kaolin clotting time (KCT), and thrombin time. The fibrinolytic effect was determined with the euglobin clot lysis time and fibrin plate methods. Extracts of green beans and tomatoes inhibited platelet aggregation induced by ADP and arachidonic acid, in a concentration-dependent manner. The methanolic extracts of grapes prolonged the PT and dPT. Finally, extracts of raspberry prolonged the APTT and also presented fibrinolytic activity. In conclusion, from a screening that included a variety of fruits and vegetables, we found antiplatelet activity in green beans and tomatoes, anticoagulant activities in grapes and raspberries, whereas fibrinolytic activity was observed only in raspberries. Further investigations are necessary to advance in knowledge of the active compounds of these fruits and vegetables and their mechanisms of action.